◇ VisionFlow

Coordination Engineering for Federated Human–AI Intelligence

Open-source coordination harness. Diverse intelligence collaborates across trust boundaries at scale.

Self-sovereign data. Formal reasoning. Cryptographic provenance. Human-in-the-loop governance.

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5	92	185+	55×	90+	6
Federated Substrates	CUDA Kernels	MCP Tools	GPU Speedup	Agent Skills	OSS Repos

PROBLEM

Hard problems share a common shape

Climate modelling, drug discovery, creative production at scale. All require diverse intelligence collaborating across trust boundaries where centralised coordination breaks down.

73% of frontline AI adoption happens without management sign-off.

Uncoordinated agents waste **60–80% of tokens** on context rediscovery, duplicate reasoning, and contradictory outputs.

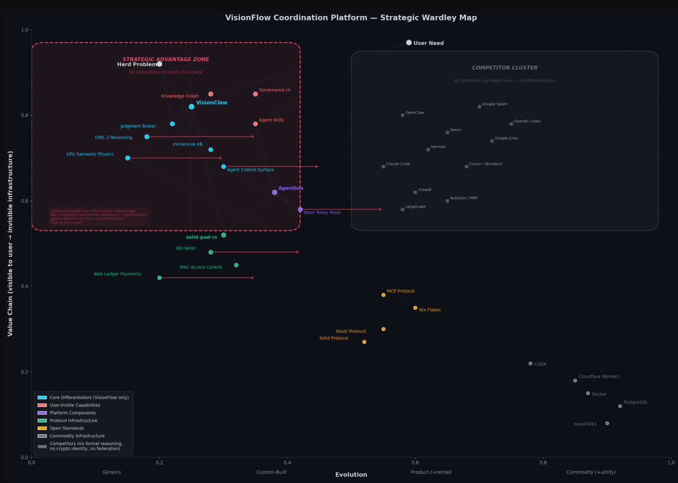
WHY TOKEN VENDORS SHOULD CARE

Coordination is the token multiplier

- **3–5× effective throughput gain.** Shared ontology eliminates re-derivation. Provenance eliminates re-validation. Governance prevents decision spin.
- **10,000:1 problem-to-token ratio.** $2.6B drug compounds justify $10K–100K in coordinated reasoning if one wrong conclusion is prevented.
- **Break-even at 3 agents.** Ontology, identity spine, governance are shared infra, not per-agent costs.
- **Model-agnostic.** Any LLM behind MCP tools. Multi-vendor consumption scales with problem complexity, not lock-in.

STRATEGIC POSITIONING: WARDLEY MAP

Every differentiator sits where no competitor can follow



VisionFlow's core differentiators (red zone, left) occupy the genesis/custom-built quadrant: OWL 2 formal reasoning, GPU semantic physics, Judgment Broker, cryptographic identity (DID:Nostr), immersive XR. No competitor has built here. All 11 competing platforms (grey cluster, right) sit in the commodity zone, interchangeable agent frameworks with no structural differentiation. The red dashed boundary marks the **strategic advantage zone**: capabilities that require deep protocol engineering. Cannot be replicated by wrapping an API.

ARCHITECTURE

Five substrates, one cryptographic identity spine

Every actor (human, agent, server) shares a single `did:nostr` secp256k1 keypair, verified at every layer.

VisionClaw | Knowledge Engineering
OWL 2 EL reasoning, 92 CUDA kernels, 55× GPU speedup, 3D graph, 17 MCP tools

Agentbox | Harness Engineering
90+ skills, 185+ MCP tools, governance relay, BIP-340 identity, Nix reproducible

solid-pod-rs | Cryptographic Foundation
Rust Solid server, DID:Nostr + WAC + Web Ledgers, NIP-98 + OIDC + WebAuthn, HTTP 402

nostr-rust-forum | Governance UI
Judgment Broker, Agent Control Surface (kinds 31400–31405), Leptos WASM, passkey auth

DreamLab Edge | Branded Deployment
React SPA + WASM forum, CF Workers edge, operator overlay, live at dreamlab-ai.com

TOKEN ECONOMICS

Tokenmaxing for hard problems is rational

Uncoordinated Agents
5–8× token waste
↓
+ Shared Ontology
Eliminate re-derivation
↓
+ Provenance Chain
Eliminate re-validation
↓
+ Judgment Broker
Eliminate decision spin
↓
Coordinated Mesh
3–5× effective throughput

HUMAN-IN-THE-LOOP GOVERNANCE

Judgment Broker

Agent publishes control panel (kind 31400) → forum renders to human → NIP-98 signed decision → VisionClaw gates knowledge mutations. Every decision is an immutable Nostr event. **90% reduction in audit reconstruction cost.**

SCALING

Single Operator | One API key, one container, minutes to deploy.

Team (50+) | Shared relay mesh, Judgment Broker oversight. Validated at scale.

Federated Enterprise | Sovereign nodes, trust via did:nostr. Horizontal scaling.

Open source. Model-agnostic. Federation-ready.

Every token produces auditable, governed, provenance-tracked reasoning.

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